Motif Evolution in Online Social Networks: An Early Virality Prediction Framework Using Temporal Graph Motifs

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***Abstract*—Online social-network discussions evolve through a sequence of interactions whose structural and temporal characteristics may provide early indications of future popularity. Existing approaches to information-cascade prediction commonly rely on aggregate cascade statistics, graph representations, or learned representations of propagation structures. However, the evolution of local interaction patterns during the earliest stage of a discussion remains relatively underexplored.**

This paper proposes a motif-evolution framework for predicting whether an online discussion will become viral using only its initial interaction history. The proposed framework observes the first 15 minutes of a discussion and represents each interaction using structural, temporal, sentiment, and reactivation-related features. Graph interactions are maintained using dynamic Compressed Sparse Row (CSR) structures, while Worst-Case Optimal Join (WCOJ) techniques are used to identify higher-order interaction motifs. The resulting event-level representation contains 20 features for each interaction and is subsequently processed as a temporal sequence using a Bidirectional Gated Recurrent Unit (BiGRU) and learned attentive pooling.

The ground-truth target is constructed independently from the complete 24-hour discussion. Rather than using a fixed global comment-count threshold, virality is defined using a subreddit-specific 95th percentile of 24-hour comment volume. This creates a leakage-controlled dataset in which early structural evolution is used to predict community-relative future popularity. The proposed framework further introduces motif completion speed and dormant-subgraph reactivation as explicit features. The paper discusses the relationship of these components to existing research on network motifs, temporal motifs, cascade prediction, graph joins, recurrent neural networks, and attention mechanisms.

***Index Terms*—online social networks, network motifs, temporal motifs, information cascades, virality prediction, graph mining, worst-case optimal joins, GRU, attention, temporal graphs**

I. Introduction

Online social networks provide an environment in which information spreads through a continuous sequence of interactions between users. A discussion that initially receives only a few comments may subsequently experience rapid growth, while another discussion with similar initial activity may become inactive. Predicting this future behaviour at an early stage is therefore an important problem in social-network analysis.

Traditional approaches to social-network analysis often represent a network as a static graph and use aggregate characteristics such as degree, density, centrality, or community structure. Although these properties provide useful information about network structure, they do not fully represent the temporal nature of online discussions. Interactions occur at specific times, and the order and rate at which users interact may contain information about the future evolution of the discussion.

Network motifs provide a useful representation for identifying recurring local structures in complex networks. Milo et al. [1] introduced network motifs as recurring patterns of interconnections that occur significantly more frequently than expected in randomized networks. Such structures can also be observed in online discussions, where replies form directed interactions between users.

However, conventional network motifs are generally concerned with structural connectivity. Temporal networks require additional information regarding the ordering and timing of interactions. Kovanen et al. [2] introduced temporal motifs to incorporate this temporal dimension into motif analysis. This provides a suitable foundation for studying how interaction structures emerge over time.

The prediction of information cascades has also been studied using machine-learning and deep-learning approaches. DeepCas, proposed by Li et al. [4], demonstrated that the structural properties of information cascades can be learned for predicting future cascade size. DeepInf, proposed by Qiu et al. [5], similarly demonstrated the usefulness of deep representation learning for social influence prediction.

Despite these developments, an important research opportunity remains in combining temporal motif evolution with very-early prediction of future discussion popularity. Instead of observing a substantially developed cascade, the proposed system restricts its predictive information to the first 15 minutes of a discussion. The subsequent 24-hour discussion is used only for generating the ground-truth virality label. This separation is intended to prevent future information from entering the predictive feature representation.

The proposed framework consists of three major processing stages. First, an online graph-processing pipeline converts incoming interactions into a dynamic directed graph and extracts structural and temporal motif features. Second, an independent target-generation pipeline analyses the complete 24-hour discussion and generates a community-relative virality label.

Finally, a neural sequence model processes the sequence of early interaction features using a Bidirectional GRU followed by learned attentive pooling.

The proposed system additionally introduces two project-specific features: motif completion speed and dormant-subgraph reactivation. Motif completion speed measures the time taken for a detected motif to develop, while dormant-subgraph reactivation captures the return of activity to previously inactive portions of the interaction graph.

The major objective of this work is therefore to investigate whether the structural and temporal evolution of interaction motifs during the earliest stage of an online discussion can provide sufficient information for predicting its future virality.

The remainder of this paper reviews relevant literature, describes the proposed dataset construction, presents the proposed methodology, and discusses the research gap and contributions of the framework.

II. Related Work

A. Network Motifs

Network motifs were introduced by Milo et al. [1] as recurring patterns of interconnections that occur more frequently than expected in randomized networks. Their work established motifs as important structural building blocks for analysing complex networks.

The same concept can be applied to online social networks by representing user interactions as directed edges. A reply from user u to user v can be represented as

$$E(u, v), \tag{1}$$

where the direction indicates the interaction relationship.

The proposed system uses several structural interaction patterns, including direct replies, sibling-star forks, directed chains, ping-pong dyads, and reciprocal cycles. These motifs provide a compact representation of the local interaction structure.

The use of network motifs itself is therefore not claimed as novel. The contribution of the proposed framework is instead the use of these motifs as event-level temporal features for early virality prediction.

B. Temporal Motifs

Kovanen et al. [2] extended conventional motif analysis by considering the temporal ordering of interactions. Temporal motifs are particularly relevant to online discussions because the same static topology may correspond to different behaviours depending on the order and timing of interactions.

For example, two users may interact repeatedly within several seconds, or the same interaction pattern may occur over several hours. A static graph would represent both cases similarly, whereas a temporal representation can distinguish them.

The proposed system therefore incorporates the time associated with each interaction. The inter-arrival time is represented as

$$\Delta t_{\text{edge}} = t_i - t_{i-1}, \tag{2}$$

where t_i is the timestamp of the current interaction.

The system additionally introduces motif completion speed:

$$\Delta t_M = t_i - t_{\text{trigger}}, \tag{3}$$

where t_{trigger} represents the earliest interaction associated with the detected motif.

While temporal information has been studied extensively in temporal network analysis, explicitly using the elapsed time required for a motif to complete as an event-level feature for early virality prediction is a distinctive aspect of the proposed framework.

C. Information Cascade Prediction

Li et al. [4] proposed DeepCas, an end-to-end neural approach for predicting future information-cascade size. The work demonstrated that structural cascade information can be transformed into learned representations for prediction.

Qiu et al. [5] proposed DeepInf for social influence prediction and demonstrated that deep learning can combine network structure and user-related information.

These studies motivate the use of learned representations for social propagation prediction. However, the proposed framework differs in that it restricts the feature-generation process to the first 15 minutes of a discussion and explicitly represents the evolution of local interaction motifs.

Consequently, the proposed task is not simply to predict the size of an already-developed cascade. Instead, it investigates whether the earliest structural evolution of the cascade can predict its future community-relative popularity.

D. Efficient Graph Pattern Matching

Efficient motif discovery becomes increasingly important as the number of interactions grows. Multi-way graph-pattern queries can produce large intermediate results when evaluated using conventional join strategies.

Veldhuizen [6] introduced Leapfrog Triejoin (LFTJ), a worst-case optimal join algorithm for evaluating multi-way joins. Its trie-based representation enables efficient intersection of relations without unnecessarily constructing large intermediate results.

The proposed framework applies Worst-Case Optimal Join techniques to motif extraction over sorted CSR adjacency structures. For example, a reciprocal interaction can be represented as

$$E(u, v) \bowtie E(v, u), \tag{4}$$

while a directed cycle can be represented as

$$E(u, v) \bowtie E(v, w) \bowtie E(w, u). \tag{5}$$

CSR and WCOJ are established techniques and are not claimed as novel algorithms in this work. Their integration into a streaming social-interaction motif extraction pipeline is instead a system-level design contribution.

E. Recurrent Neural Networks and GRU

Cho et al. [7] introduced the Gated Recurrent Unit (GRU) as a recurrent architecture for modelling sequential dependencies. The gating mechanism enables the model to retain relevant information while processing a sequence.

The proposed system represents a discussion as a sequence

$$X_i \in \mathbb{R}^{K_i \times 20}, \tag{6}$$

where each row corresponds to an interaction and contains structural, temporal, sentiment, and reactivation-related features.

A Bidirectional GRU is then used to generate hidden representations:

$$H = \mathrm{BiGRU}(X). \tag{7}$$

The GRU architecture itself is established. The project-specific aspect is the application of the model to a motif-centric temporal representation rather than directly processing raw text or only aggregate cascade statistics.

F. Attention Mechanisms

Attention mechanisms allow a model to assign different importance to different elements of a sequence. Luong et al. [8] demonstrated the effectiveness of attention mechanisms for sequence modelling.

The proposed model applies learned attentive pooling to the hidden states generated by the Bidirectional GRU. The attention score for event j is calculated as

$$s_j = w_2^T \tanh(W_1 h_j + b_1) + b_2. \tag{8}$$

The attention weights are calculated using

$$\alpha_j = \frac{\exp(s_j)}{\sum_{k=1}^{K} \exp(s_k)}. \tag{9}$$

The final representation is

$$c = \sum_{j=1}^{K} \alpha_j h_j. \tag{10}$$

Attention is not itself a novel contribution. However, applying attention over motif-centric interaction events enables the model to learn which early structural events contribute most strongly to future virality.

III. Proposed Dataset Construction

A. Temporal Separation of Features and Labels

A central characteristic of the proposed dataset is the separation between the feature-generation horizon and the target-generation horizon.

Only interactions occurring during the first 15 minutes following the creation of a post are used to generate predictive features. The full 24-hour interaction history is subsequently used to construct the ground-truth label.

This can be represented as

$$\underbrace{[0, 15 \text{ min}]}_{\text{Feature horizon}} \rightarrow X, \tag{11}$$

while

$$\underbrace{[0, 24 \text{ h}]}_{\text{Target horizon}} \rightarrow y. \tag{12}$$

The separation prevents interactions occurring after the prediction window from becoming predictive features.

B. Event-Level Dataset Representation

Instead of representing each discussion using a single fixed-length vector, the proposed dataset retains the temporal sequence of interactions.

For discussion i, the feature representation is

$$X_i = [x_1, x_2, \ldots, x_{K_i}]^T, \tag{13}$$

where

$$X_i \in \mathbb{R}^{K_i \times 20}. \tag{14}$$

The number of interactions K_i may vary between discussions.

This representation allows the neural model to observe the evolution of the discussion rather than only its final state within the 15-minute observation window.

C. Community-Relative Virality Label

A fixed global comment threshold is potentially unsuitable because different communities exhibit substantially different levels of activity.

The proposed system therefore calculates a subreddit-specific 95th-percentile threshold:

$$C_s = P_{95}(\{C_{24h} \mid post \in s\}), \tag{15}$$

where C_{24h} represents the number of comments received during the 24-hour period.

The binary target is

$$y_{\text{viral}} = \begin{cases} 1, & C_{24h} \geq C_s, \\ 0, & C_{24h} < C_s. \end{cases} \tag{16}$$

A minimum activity threshold is additionally used to avoid unstable labels for communities with very small numbers of observations.

This dataset construction is a potentially distinctive aspect of the project because the target represents exceptional activity relative to the corresponding community rather than absolute popularity.

However, the use of percentile-based thresholds itself is not claimed as universally novel without systematic comparison against existing datasets.

D. Novelty of Dataset Formation

The dataset formation can best be described as a **novel system-level formulation**.

The individual components of the dataset are based on established concepts. However, the proposed dataset combines:

1) a strict 15-minute observation window;
2) event-level temporal motif representation;
3) motif completion speed;
4) dormant-subgraph reactivation;
5) sentiment information;
6) graph structural information;
7) a subreddit-specific 95th-percentile target; and
8) a 24-hour prediction horizon.

Thus, the potential novelty lies primarily in the combination and organization of these components into a leakage-controlled dataset for very-early virality prediction.

IV. Proposed Methodology

A. Dynamic Graph Representation

Each comment interaction is represented as a directed graph edge

$$E(u, v, t), \tag{17}$$

where u is the author of the comment, v is the parent-comment author, and t is the timestamp.

The graph is maintained using dynamic CSR structures containing adjacency indices and timestamp information. This representation provides efficient access to the neighborhood of a node.

B. Motif Extraction

The proposed system identifies several interaction motifs, including:

- direct replies;
- sibling-star forks;
- directed chains;
- ping-pong dyads;
- reciprocal cycles.

The motif type is encoded using a one-hot representation.

For example, a reciprocal interaction is represented by

$$E(u, v) \wedge E(v, u). \tag{18}$$

A three-node directed cycle is represented as

$$E(u, v) \wedge E(v, w) \wedge E(w, u). \tag{19}$$

The motif engine uses WCOJ-style operations over sorted adjacency lists.

C. Structural Features

For each interaction, structural properties are extracted from the current graph state. These include measures such as:

- tree depth;
- parent out-degree;
- CSR node count;
- distance from the discussion root.

These features describe the local position of an interaction within the evolving discussion graph.

D. Temporal Features

Temporal information is represented using both inter-arrival time and motif completion speed.

The inter-arrival time is

$$\Delta t_{\text{edge}} = t_i - t_{i-1}. \tag{20}$$

The motif completion speed is

$$\Delta t_M = t_i - t_{\text{trigger}}. \tag{21}$$

The second feature is particularly relevant to the proposed research because it captures the rate at which an interaction structure develops.

E. Dormant-Subgraph Reactivation

The system determines whether an interaction reactivates a previously inactive portion of the graph.

For node u,

$$T_{\text{idle}} = t_i - \max(t_{\text{previous}}(u)). \tag{22}$$

The node is considered reactivated when

$$T_{\text{idle}} > T_{\text{threshold}}. \tag{23}$$

The representation includes:

- reactivation indicator;
- idle duration;
- number of reactivated descendants.

This is one of the more distinctive proposed features because it captures the return of activity to dormant graph structure.

F. 20-Dimensional Feature Vector

Each interaction is represented by a 20-dimensional feature vector:

$$x_i \in \mathbb{R}^{20}. \tag{24}$$

The feature categories include:

For a discussion containing K interactions,

$$X = [x_1, x_2, \ldots, x_K]^T \in \mathbb{R}^{K \times 20}. \tag{25}$$

TABLE I
FEATURE CATEGORIES IN THE PROPOSED REPRESENTATION

Category	Information
Motif	Motif-type indicators
Structure	Graph structural properties
Sentiment	Sentiment and polarity
Reactivation	Idle and reactivation information
Temporal	Motif speed and inter-arrival time
Context	Window and cumulative activity

G. Bidirectional GRU

The sequence X is passed to a Bidirectional GRU:

$$H = [h_1, h_2, \ldots, h_K]. \tag{26}$$

The forward and backward hidden states allow each event representation to incorporate information from both directions within the observed sequence.

H. Attentive Pooling

The hidden states are aggregated using learned attention. The attention score is

$$s_j = w_2^T \tanh(W_1 h_j + b_1) + b_2. \tag{27}$$

The attention coefficient is

$$\alpha_j = \frac{\exp(s_j)}{\sum_{k=1}^{K} \exp(s_k)}. \tag{28}$$

The sequence is converted into a fixed-length representation:

$$c = \sum_{j=1}^{K} \alpha_j h_j. \tag{29}$$

The resulting vector is passed to an MLP classifier to obtain the probability of virality.

V. Research Gap and Contributions

The literature establishes strong foundations for network motifs, temporal networks, cascade prediction, graph-pattern processing, and neural sequence modelling. Nevertheless, these areas are generally studied as separate components.

The proposed work attempts to bridge these areas through a unified early-prediction framework.

The principal contributions of the proposed system are:

1) **Early temporal motif prediction:** The framework investigates whether interaction motifs observed during only the first 15 minutes can predict 24-hour virality.
2) **Motif completion speed:** The time required for a motif to emerge is explicitly represented as a predictive feature.
3) **Dormant-subgraph reactivation:** The framework captures the reactivation of previously inactive portions of the interaction graph.
4) **Event-level 20-D representation:** Structural, temporal, sentiment, contextual, and reactivation information are combined into a single interaction-level vector.
5) **Community-relative target construction:** The ground truth is constructed using a subreddit-specific 95th-percentile threshold rather than a fixed global threshold.
6) **Integrated graph and neural architecture:** WCOJ-based motif extraction is connected to a BiGRU and attention model through a common temporal feature representation.

The novelty of the proposed work should therefore be understood primarily as a *system-level and feature-level contribution*. The underlying components such as CSR, temporal motifs, WCOJ, GRU, attention, and sentiment analysis are established techniques.

VI. Discussion

The proposed framework provides a different perspective on early virality prediction. Rather than treating a discussion as a single static graph, it represents the discussion as a sequence of evolving local structures.

This distinction is important because two discussions may have the same number of comments and similar final graph statistics while exhibiting very different early interaction dynamics. One discussion may rapidly form reciprocal interactions and activate previously inactive regions, whereas another may consist primarily of isolated direct replies.

The proposed representation allows these differences to be retained. Motif identity describes the structural pattern, temporal features describe its rate of development, reactivation features describe changes in activity, and sentiment provides complementary information about the content of interactions.

The Bidirectional GRU then provides a mechanism for learning temporal dependencies among these events, while attention provides a mechanism for assigning different importance to individual events.

An important advantage of the proposed dataset formulation is the separation between feature generation and target generation. Since only the first 15 minutes are used as predictive input, the model can be evaluated as an actual early-warning system rather than as a model that observes a large portion of the cascade before making its prediction.

VII. Conclusion

This paper presented a proposed framework for early virality prediction in online social networks using temporal motif evolution. The framework combines dynamic graph representation, motif extraction, temporal features, dormant-subgraph reactivation, sentiment information, and neural sequence modelling.

The proposed dataset construction separates the first 15 minutes of predictive information from the 24-hour ground-truth horizon. It represents each early interaction using a 20-dimensional feature vector and defines virality relative to the activity distribution of the corresponding subreddit.

The most distinctive aspects of the proposed system are the explicit representation of motif completion speed and dormant-subgraph reactivation, together with their integration into an event-level motif representation. The use of WCOJ over CSR structures provides an efficient graph-processing mechanism, while the Bidirectional GRU and attention mechanism provide a means of learning temporal dependencies and identifying important early interactions.

The underlying algorithms used by the system are established methods. Consequently, the primary contribution of the work is the integration of these techniques into a unified framework for predicting community-relative virality from the earliest evolution of an online discussion.

Future evaluation should include ablation studies for motif features, reactivation features, temporal features, sentiment, attentive pooling, and the 15-minute observation window. Such experiments would help determine which components provide the greatest predictive contribution and whether the proposed motif-evolution representation provides measurable improvements over conventional cascade features.

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